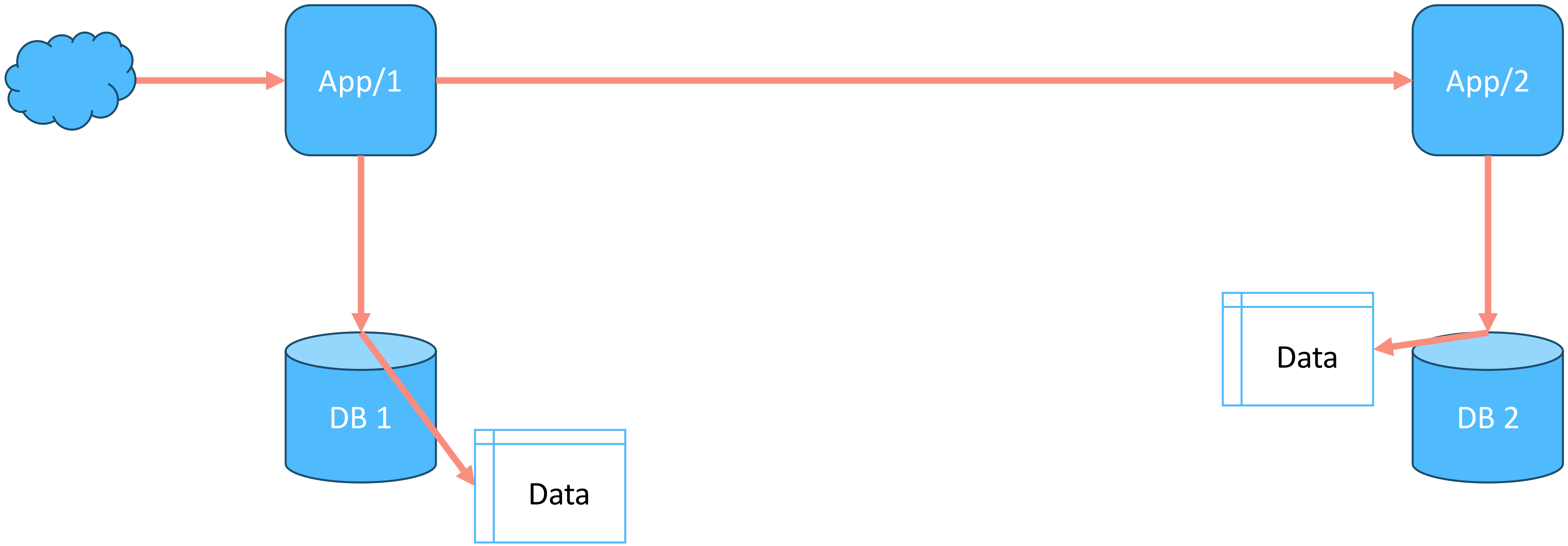
Queues Crash Course

Transactional Outbox

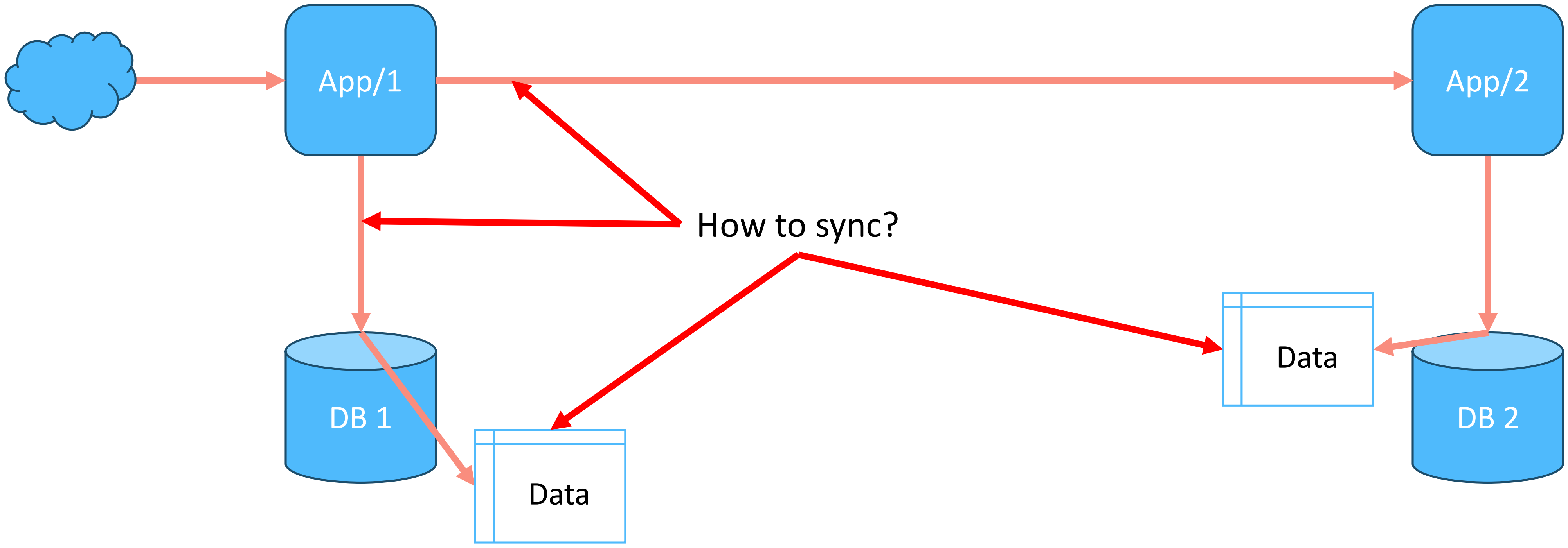
Transactional Inbox

CDC (Change Data Capture)

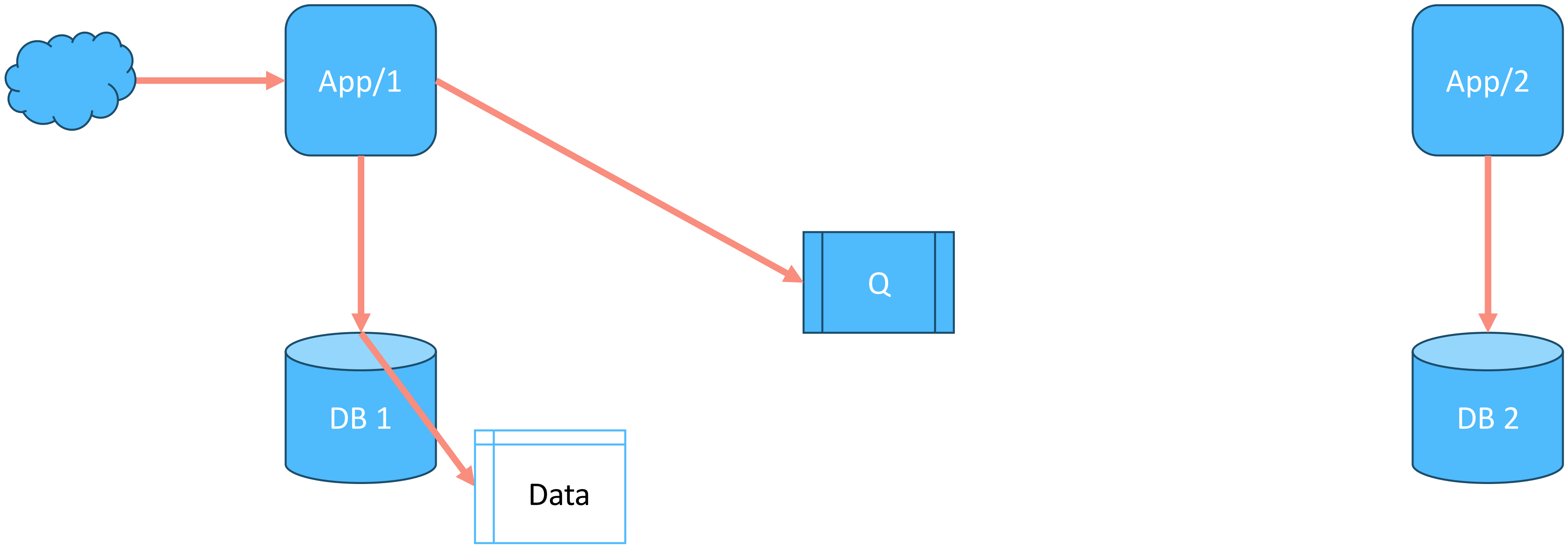
Distributed databases, same state



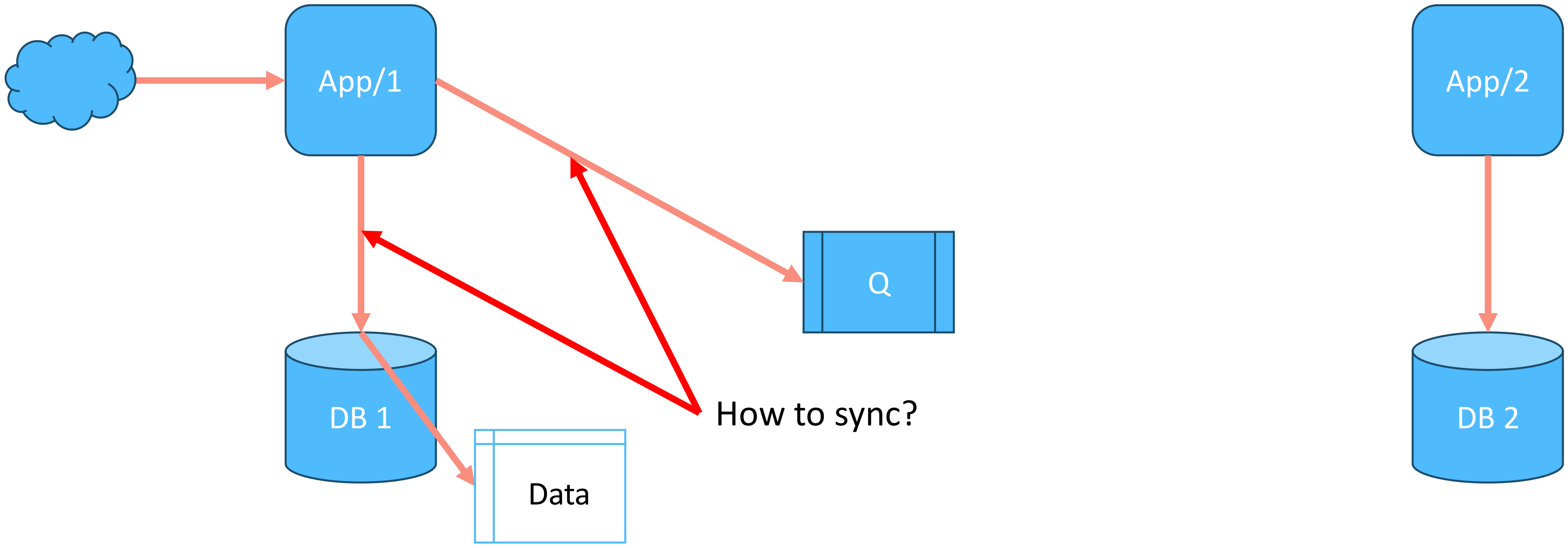
Distributed databases, same state



Transactional outbox



Transactional outbox



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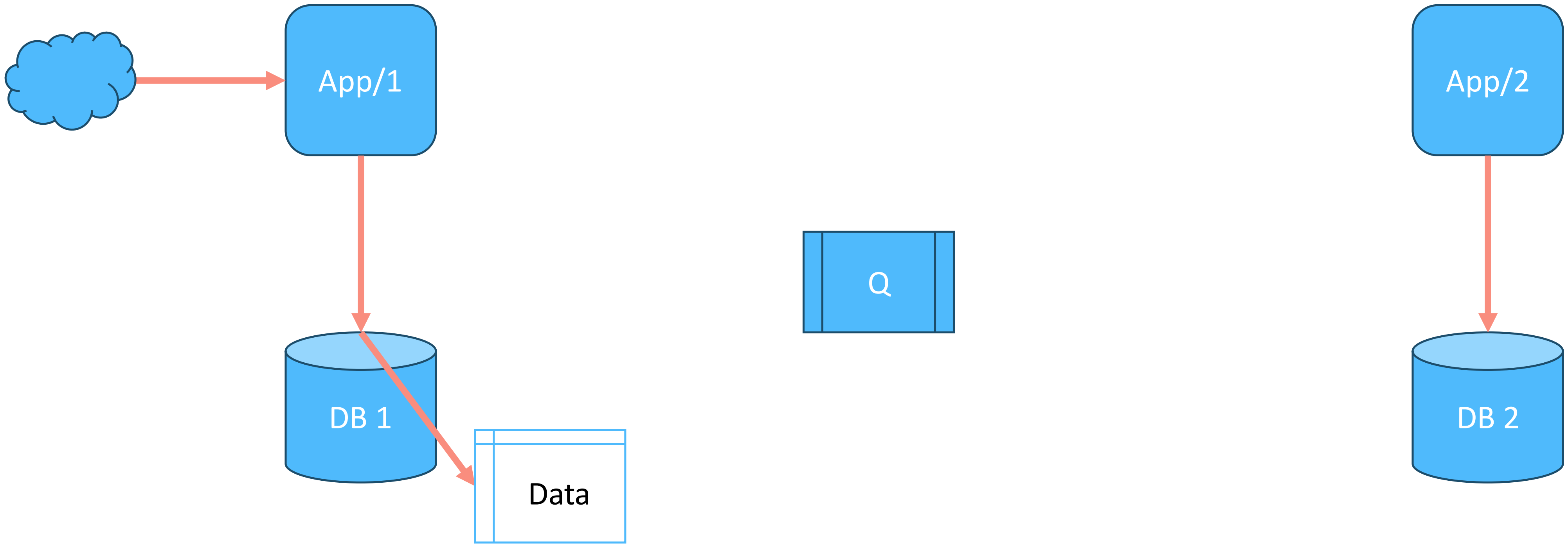

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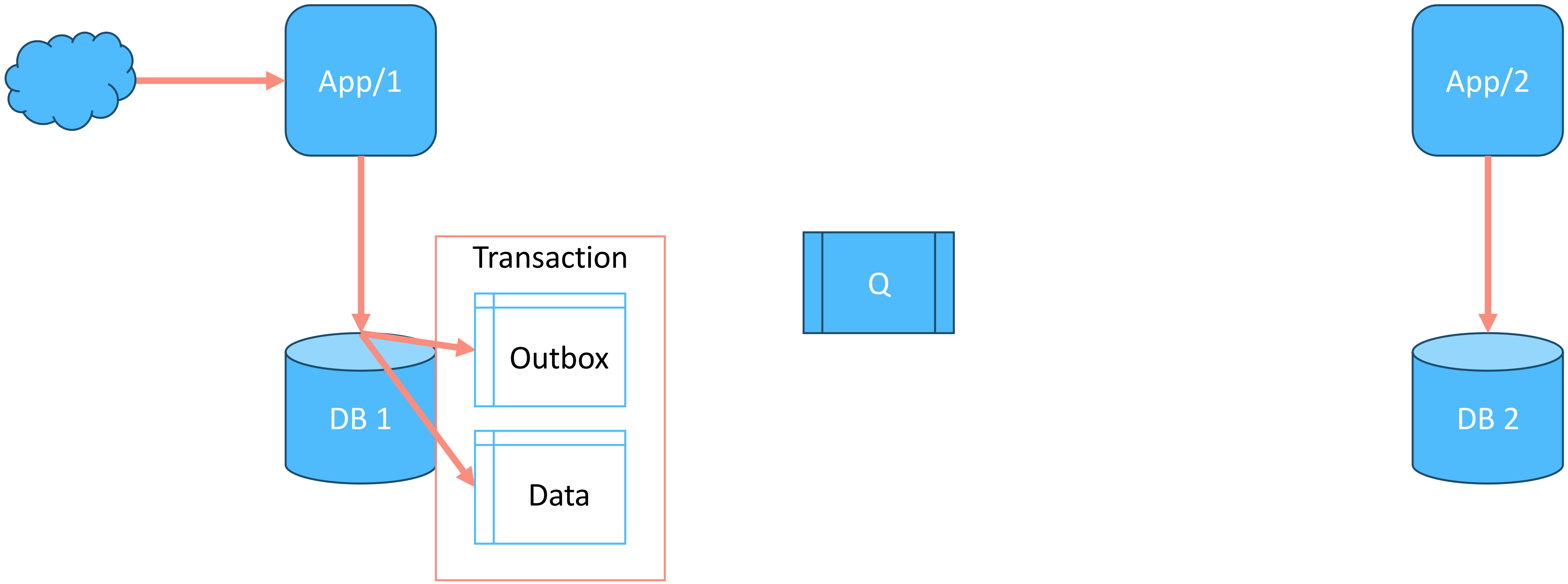

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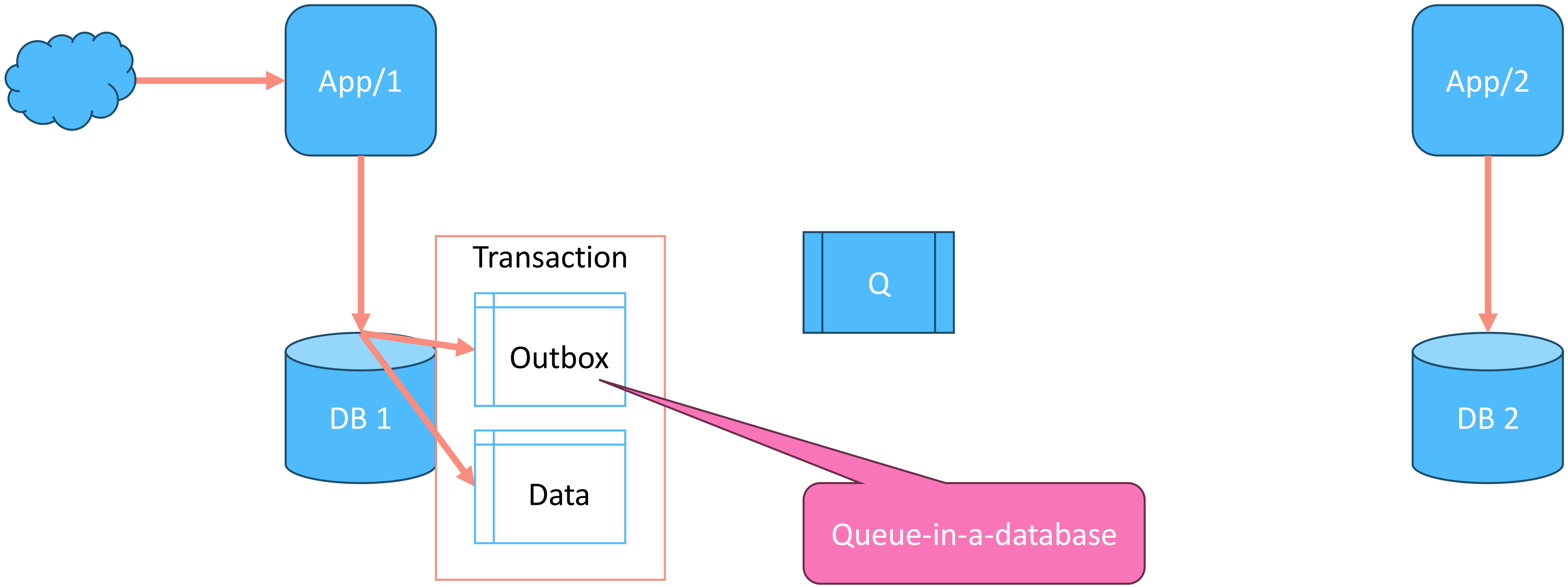

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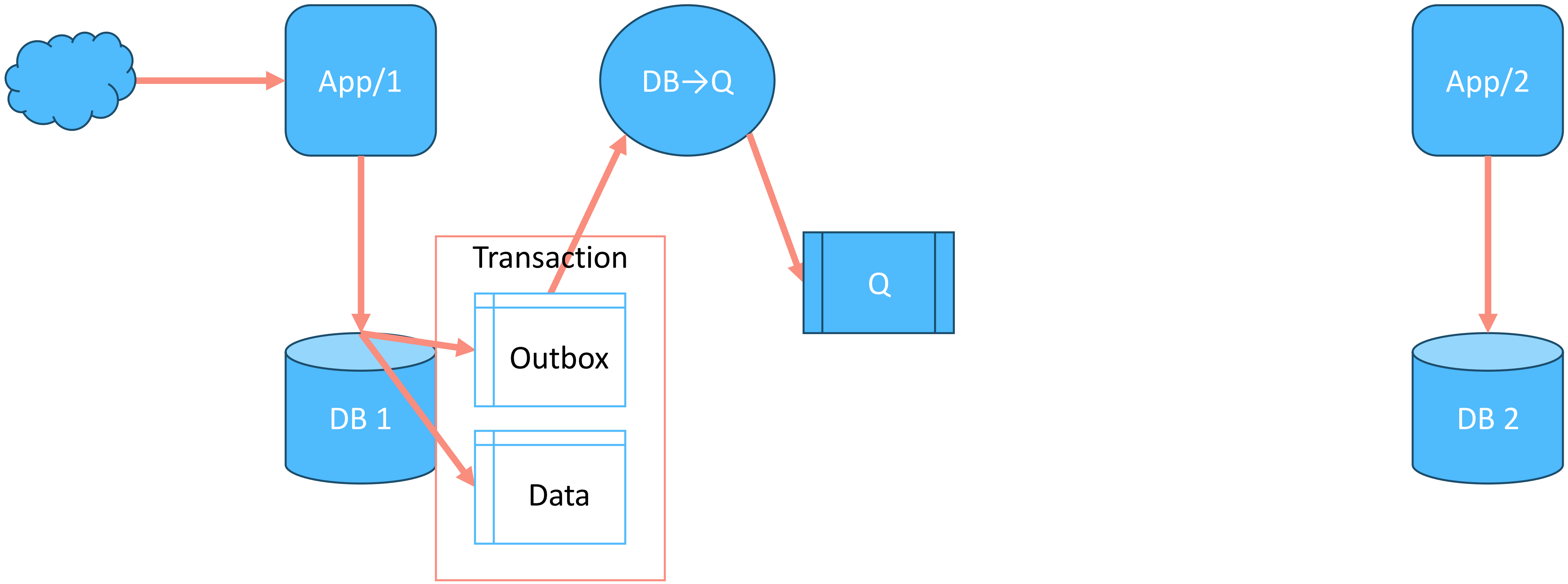

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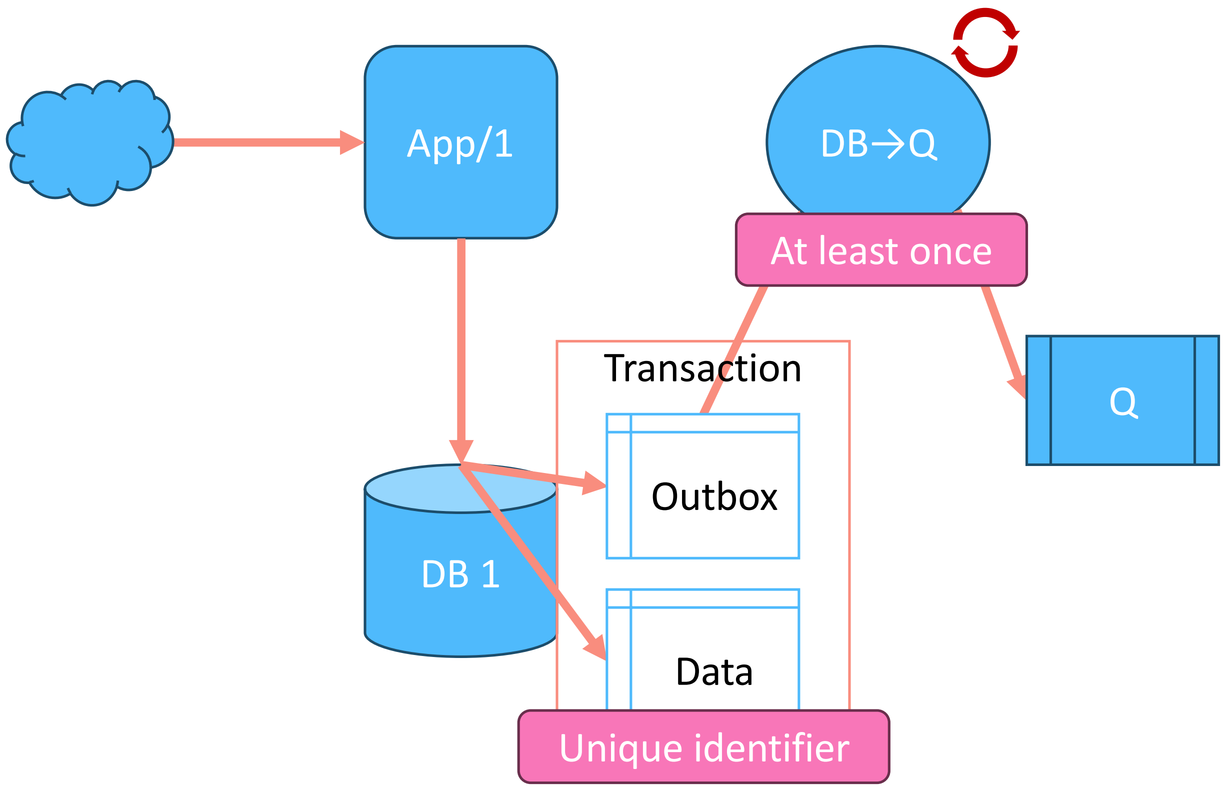
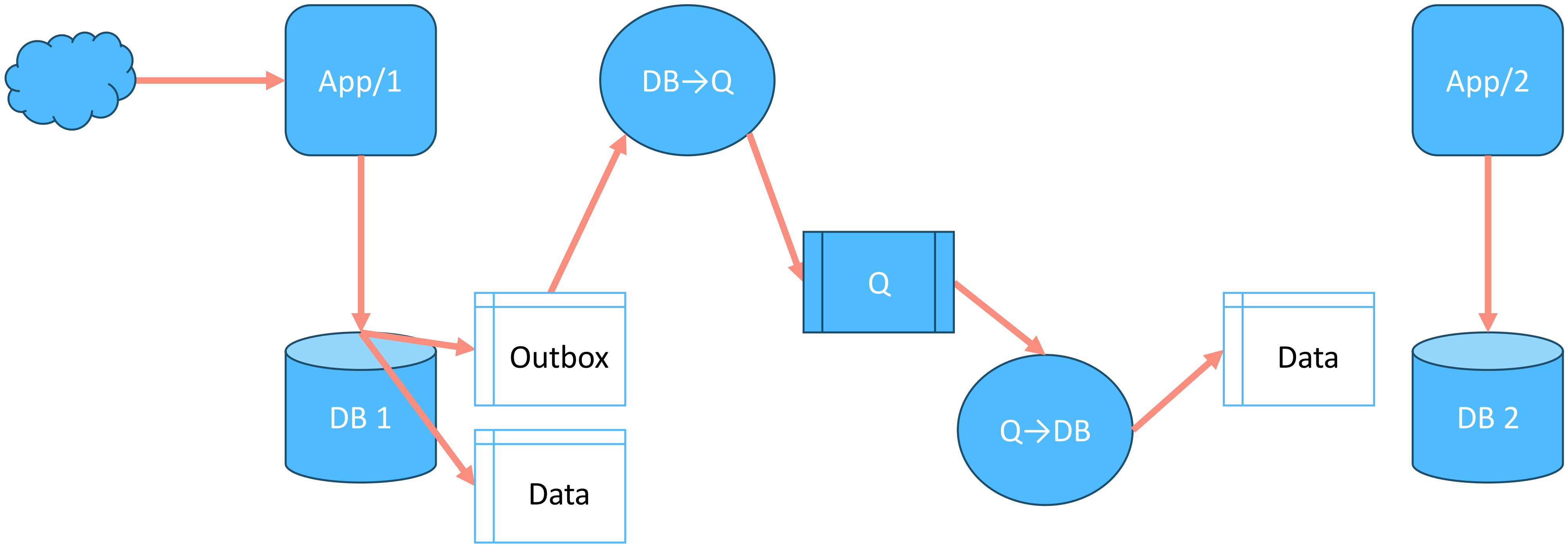


Transactional inbox



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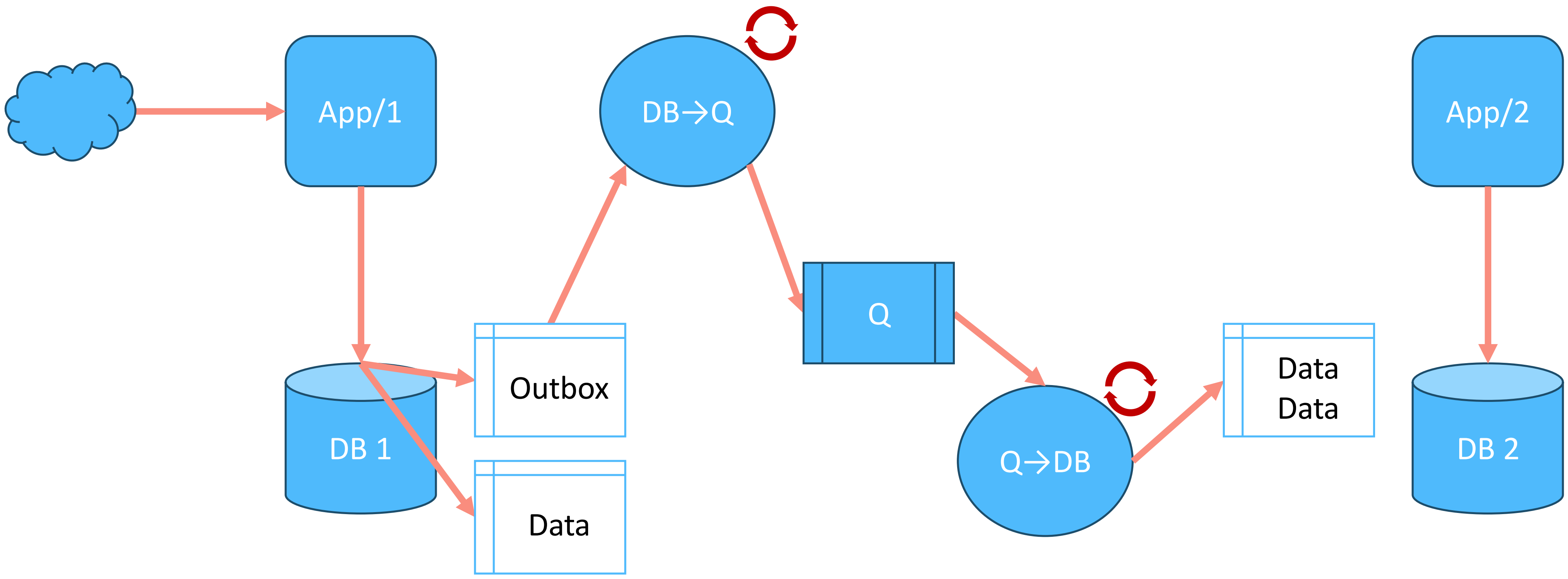
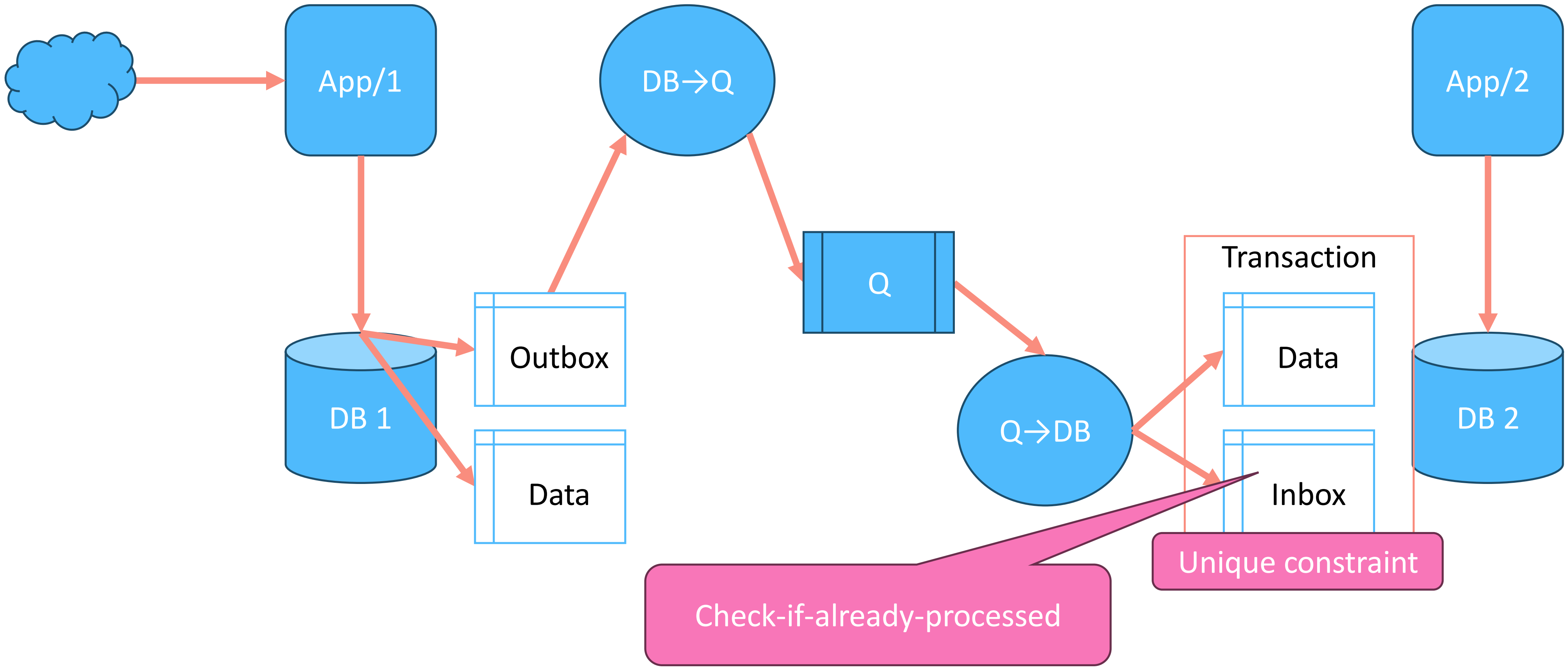


Transactional inbox



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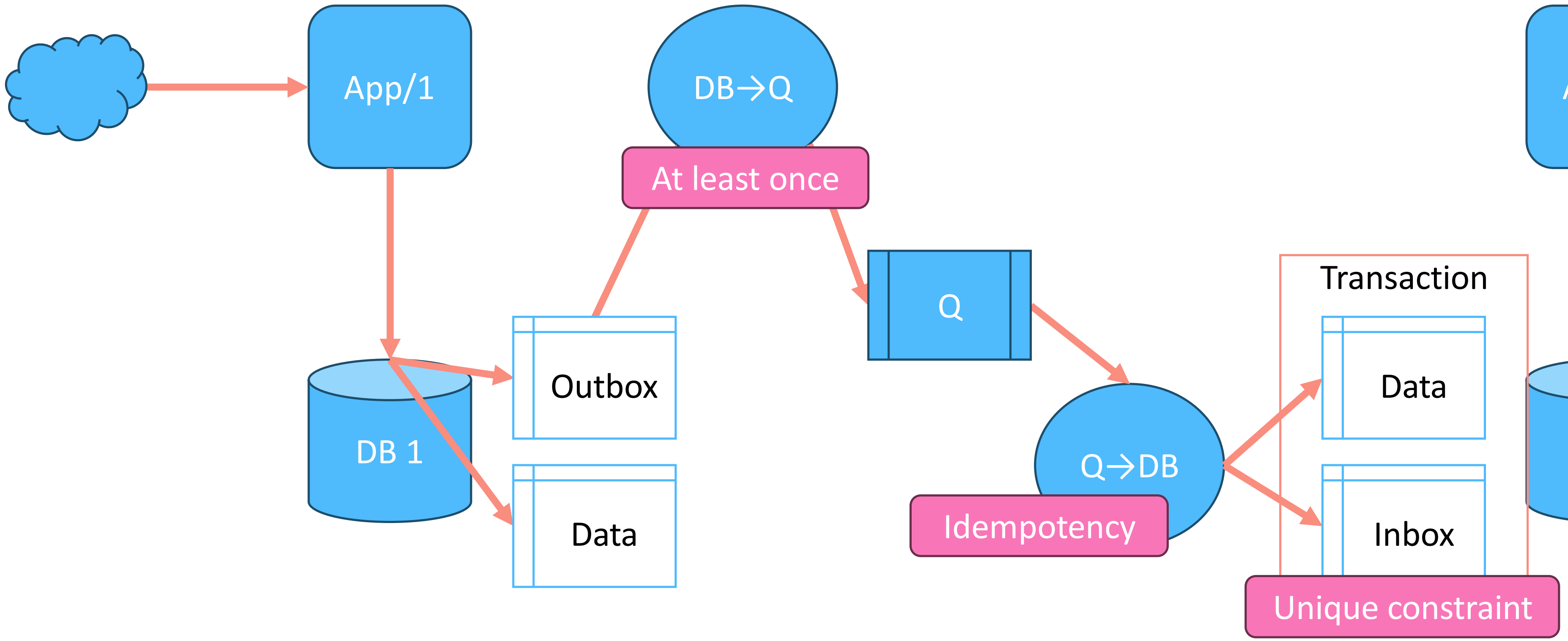
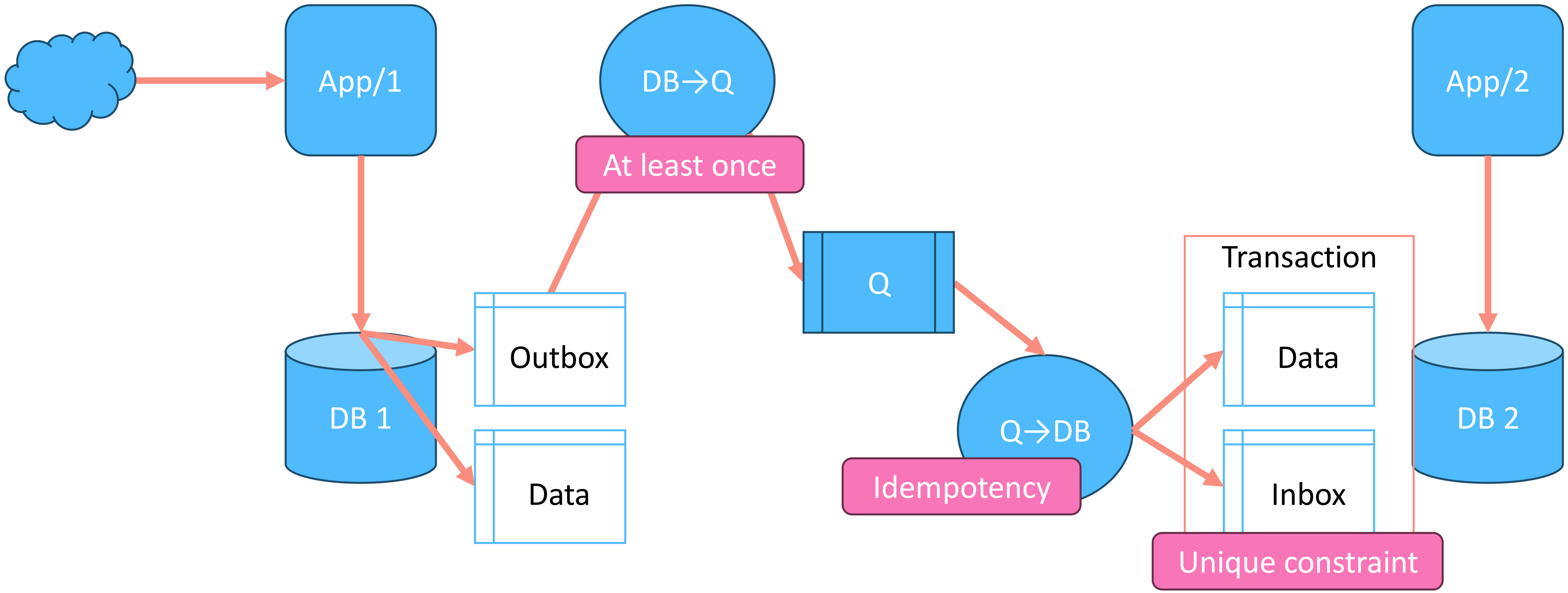


CDC: Change Data Capture



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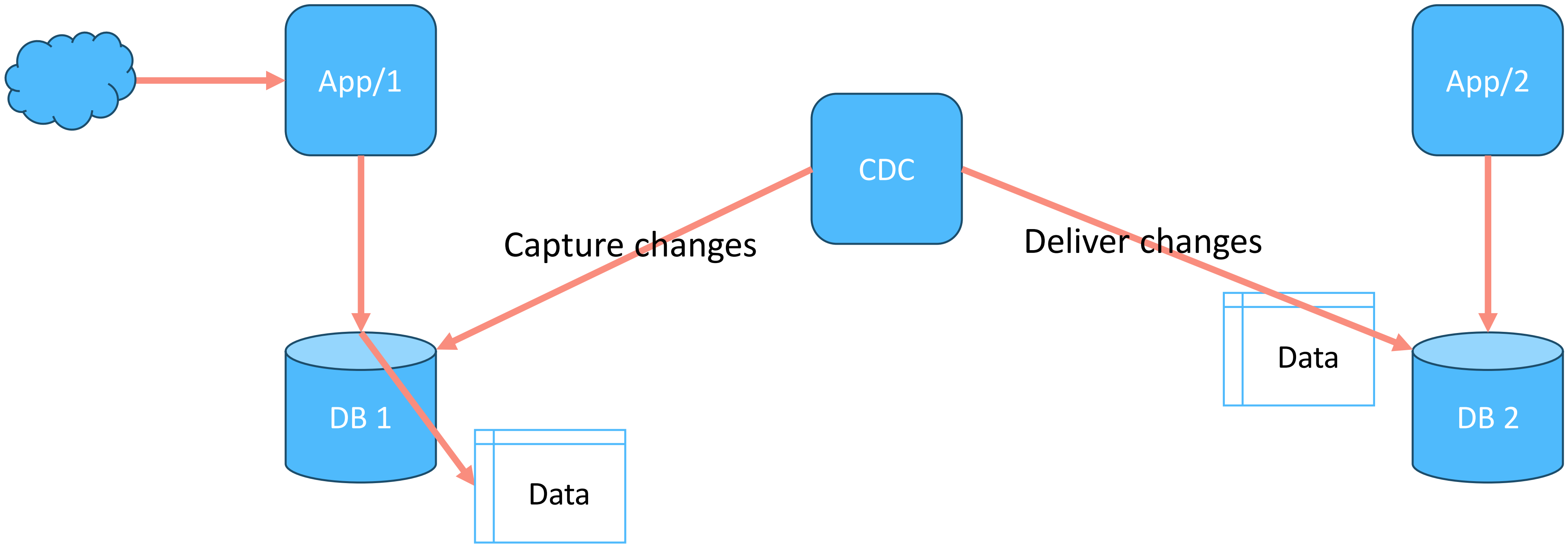

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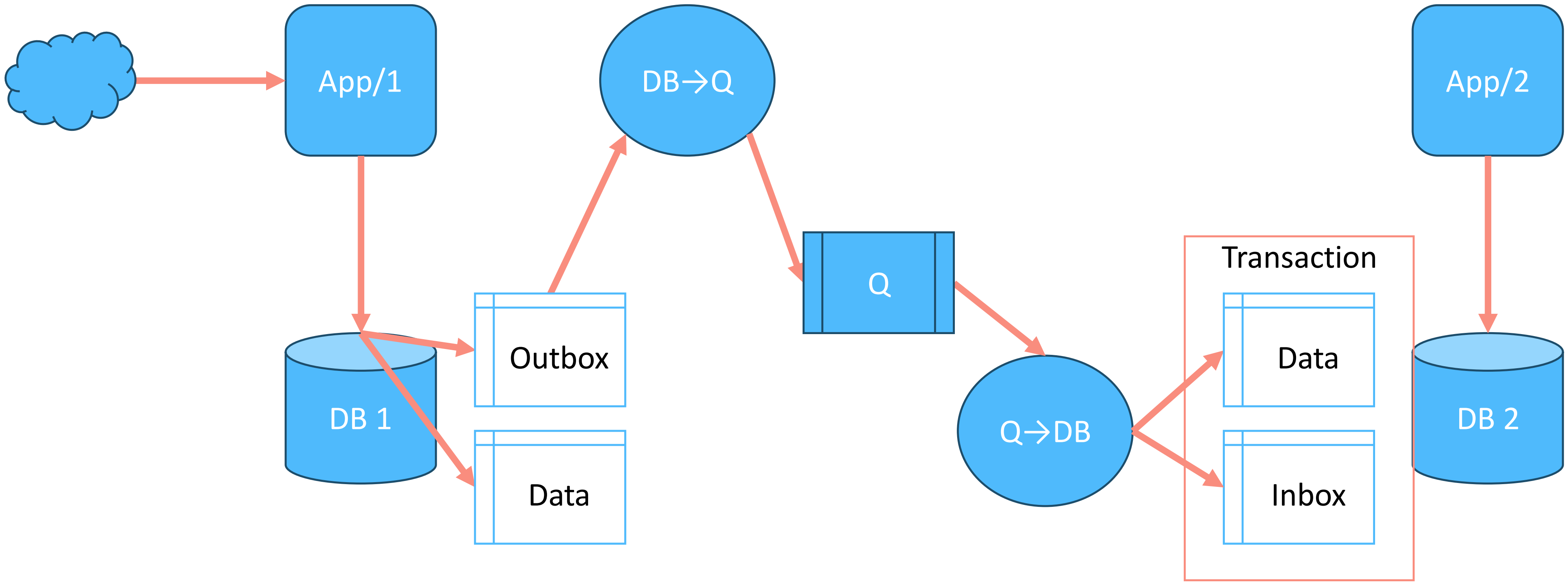

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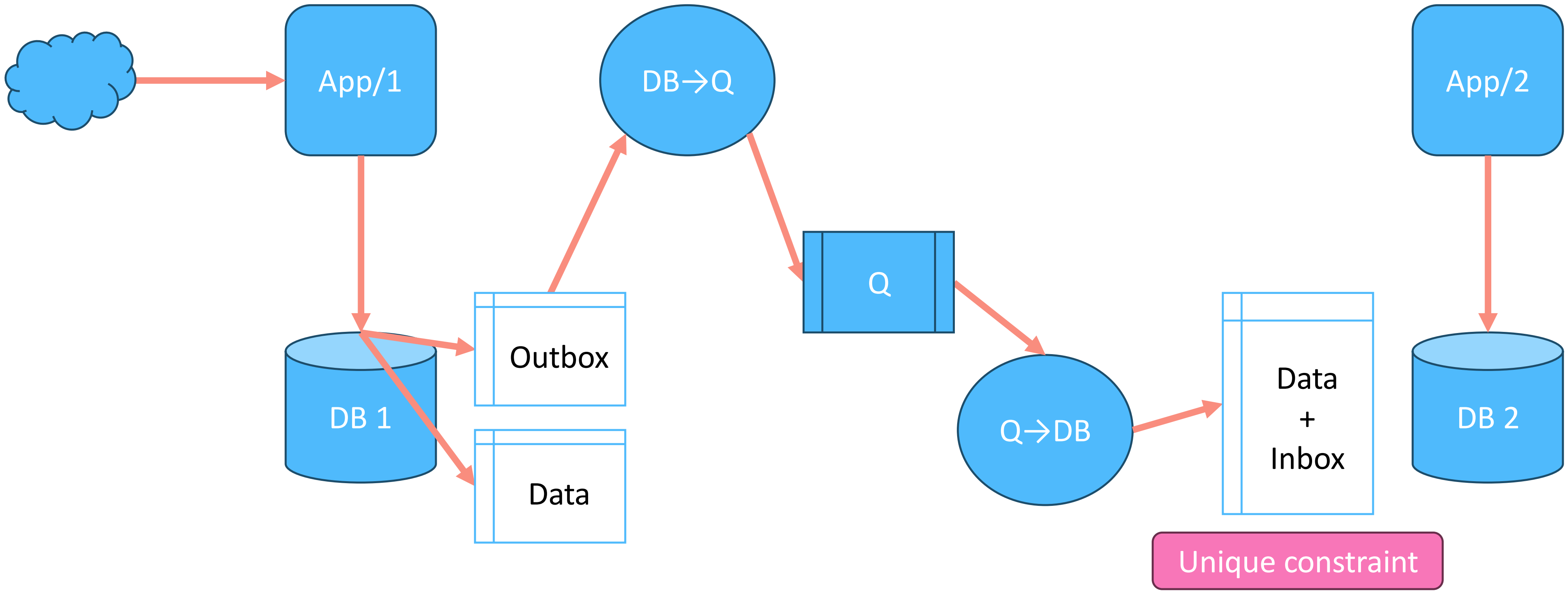
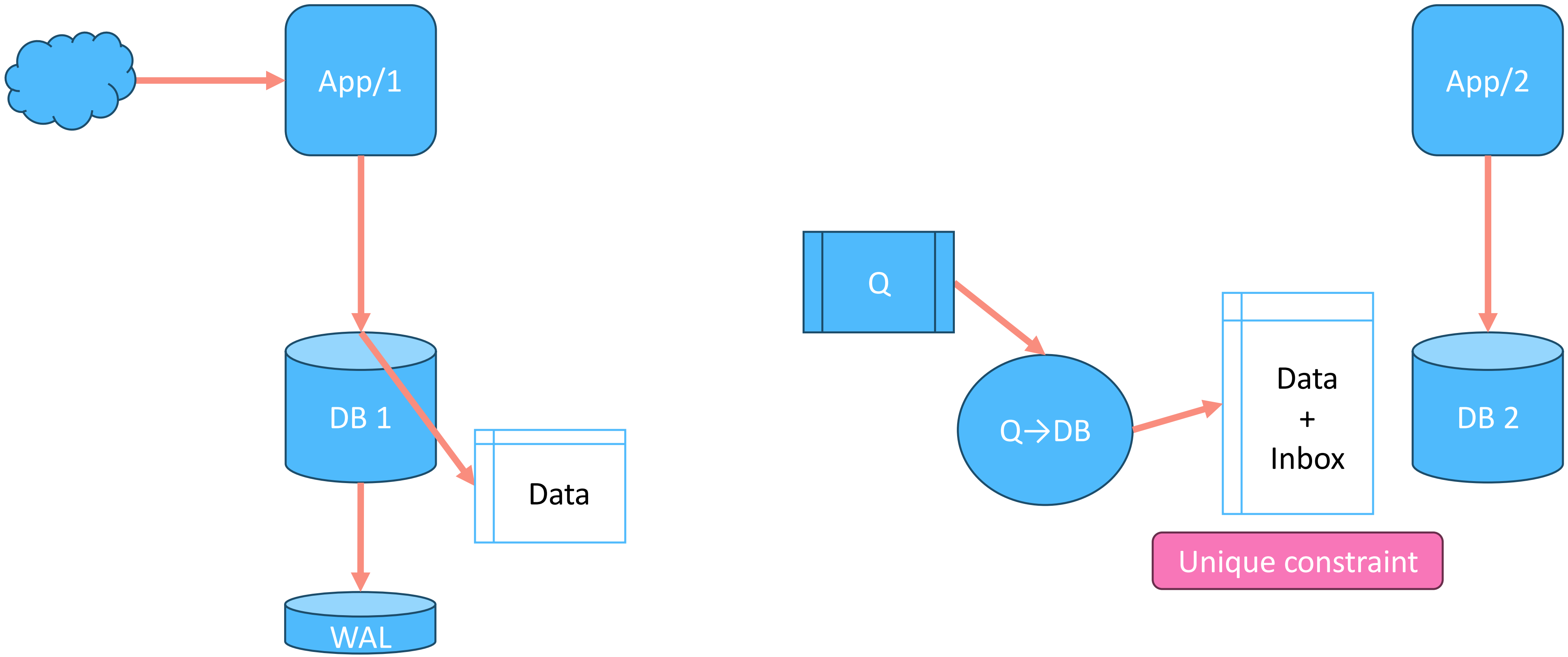


Capture changes, no intrusion



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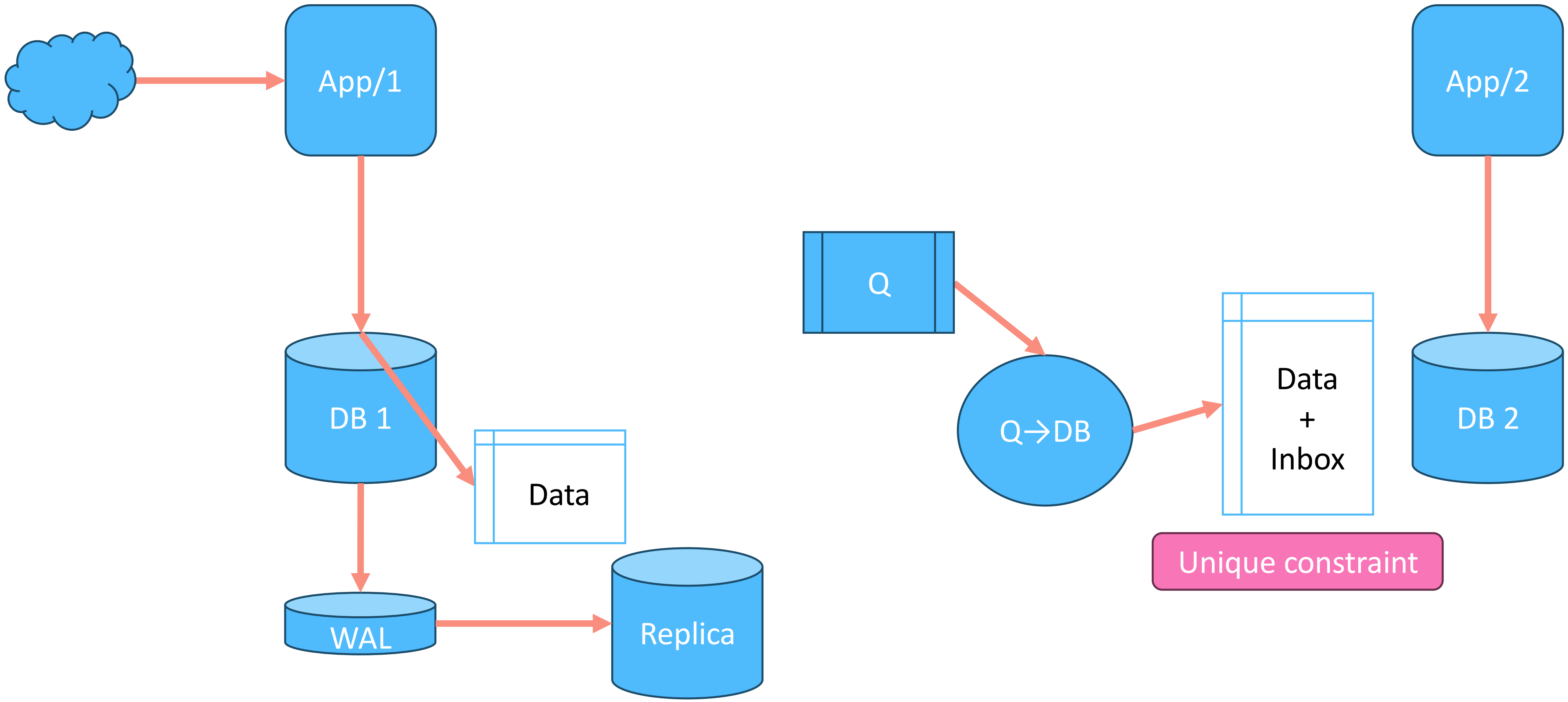

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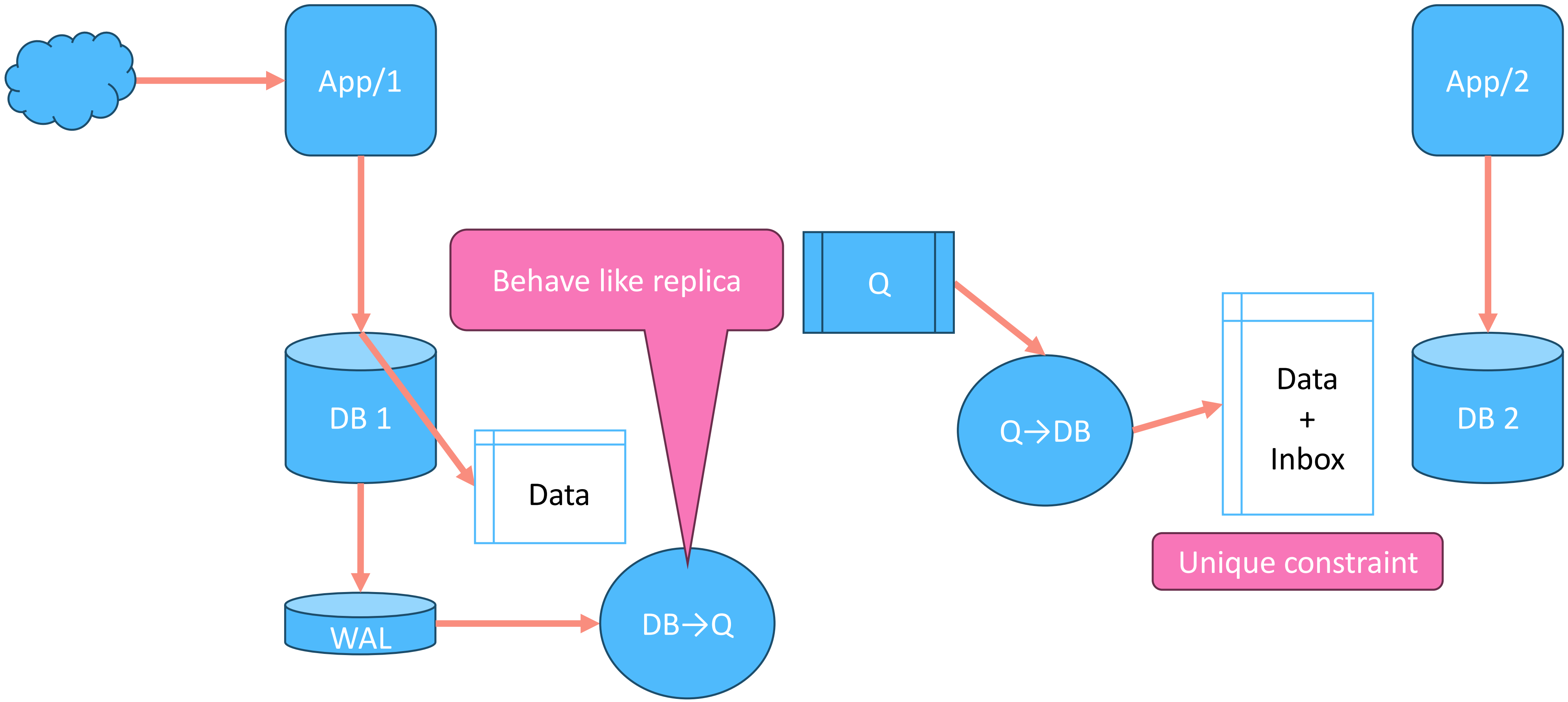

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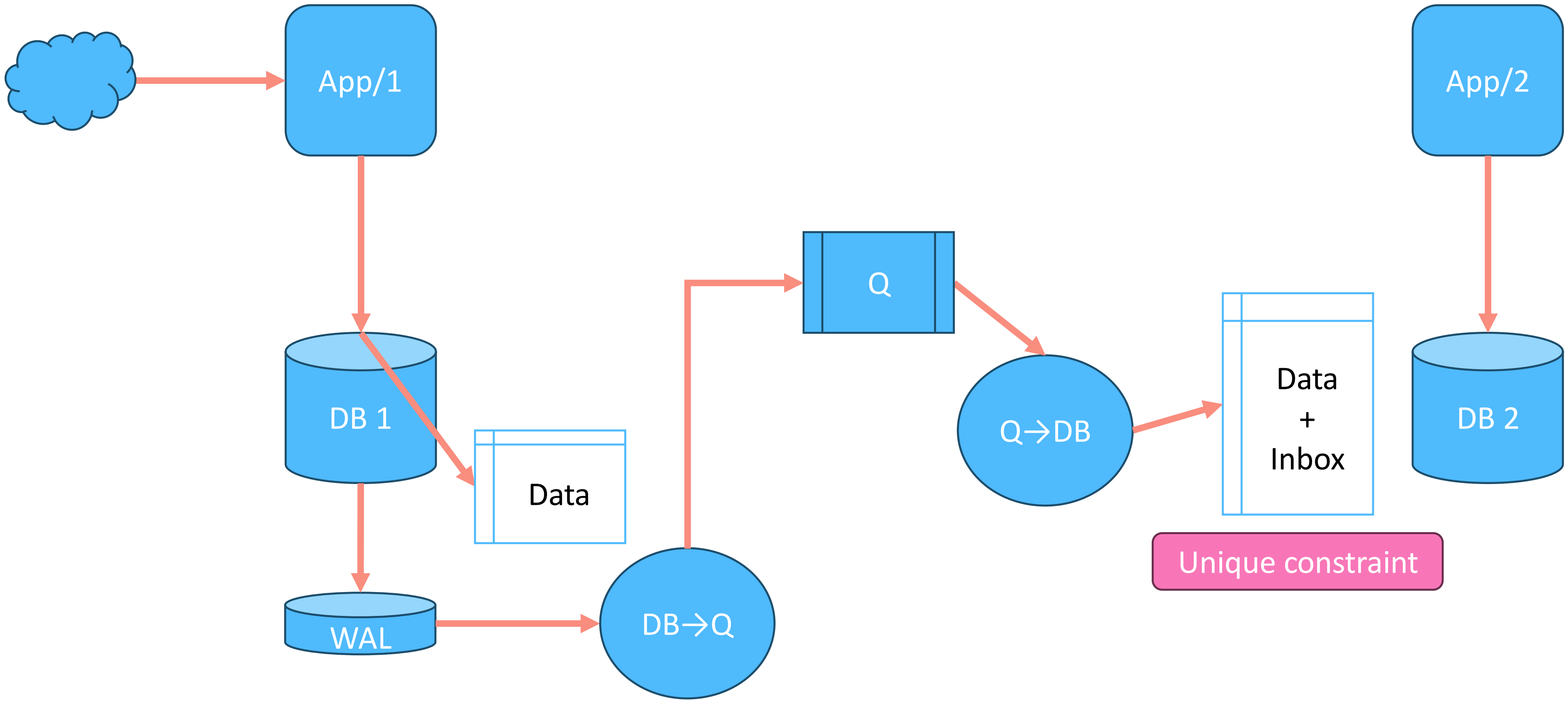
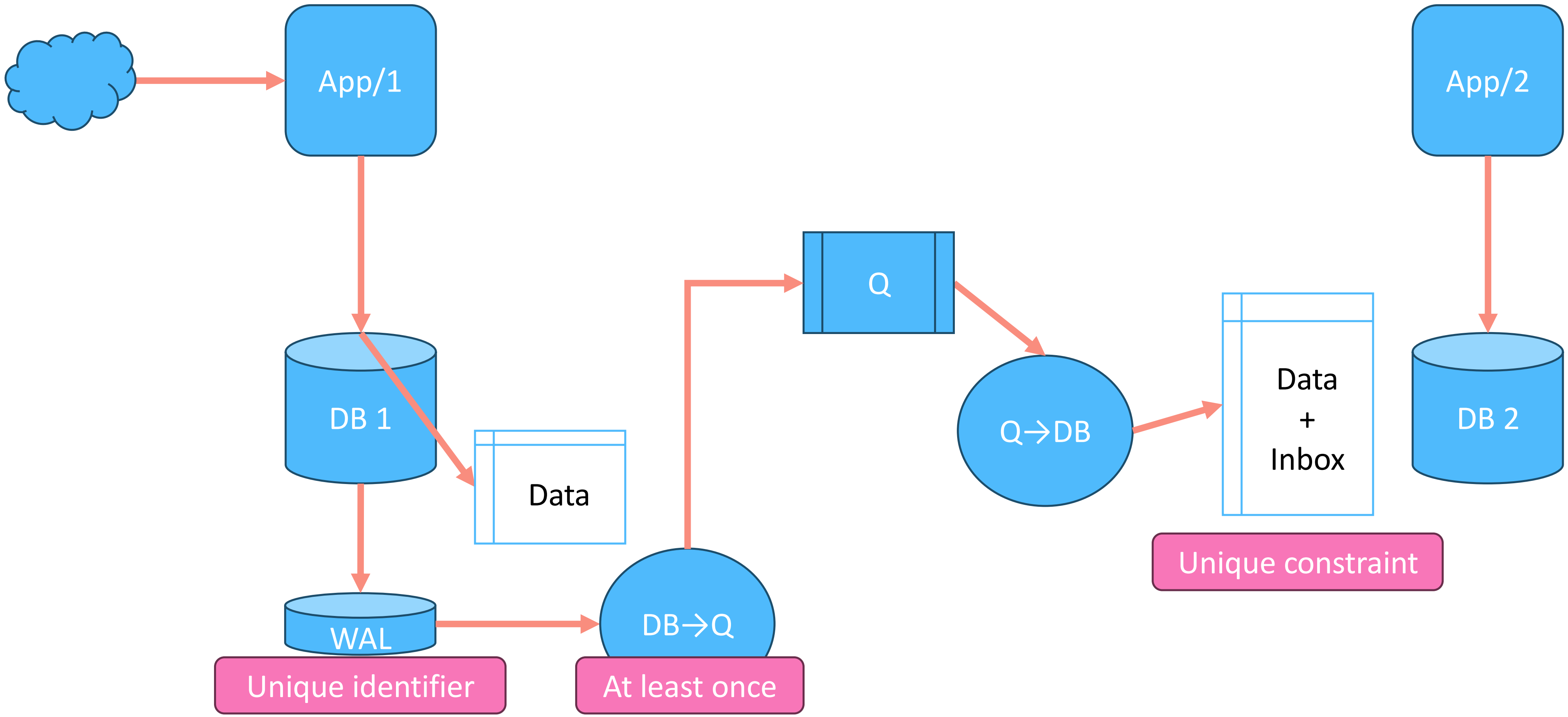


CDC from WAL


App/1
DB 1
Data
WAL
Unique identifier
DB→Q
At least once
Q
Q→DB
Data
+
Inbox
Unique constraint
App/2
DB 2

CDC Approaches

- Timestamps/version/status on rows (major intrusion)
  - Requires schema modifications
  - Alteration of application
  - Additional database load
- Triggers on tables (minor intrusion)
  - Opaque for the application
  - Additional database load
- Capture replication/wal (no intrusion)
  - No additional database load
  - No effect on schema or application

Further reading

Further reading

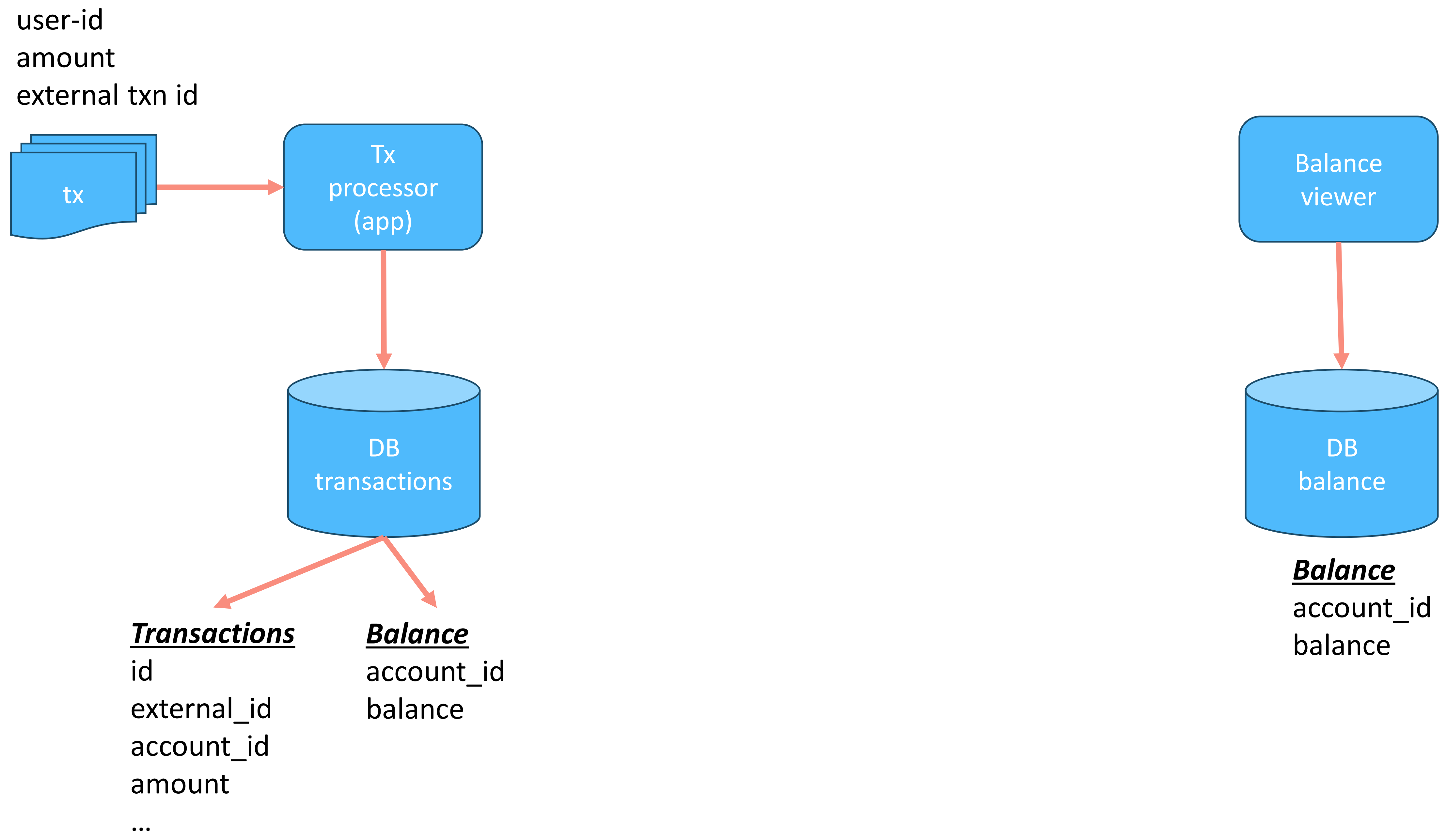
- https://softwaremill.com/microservices-101/
- https://microservices.io/patterns/data/transactional-outbox.html
- https://microservices.io/patterns/data/transaction-log-tailing.html
- https://en.wikipedia.org/wiki/Inbox_and_outbox_pattern
- https://debezium.io/

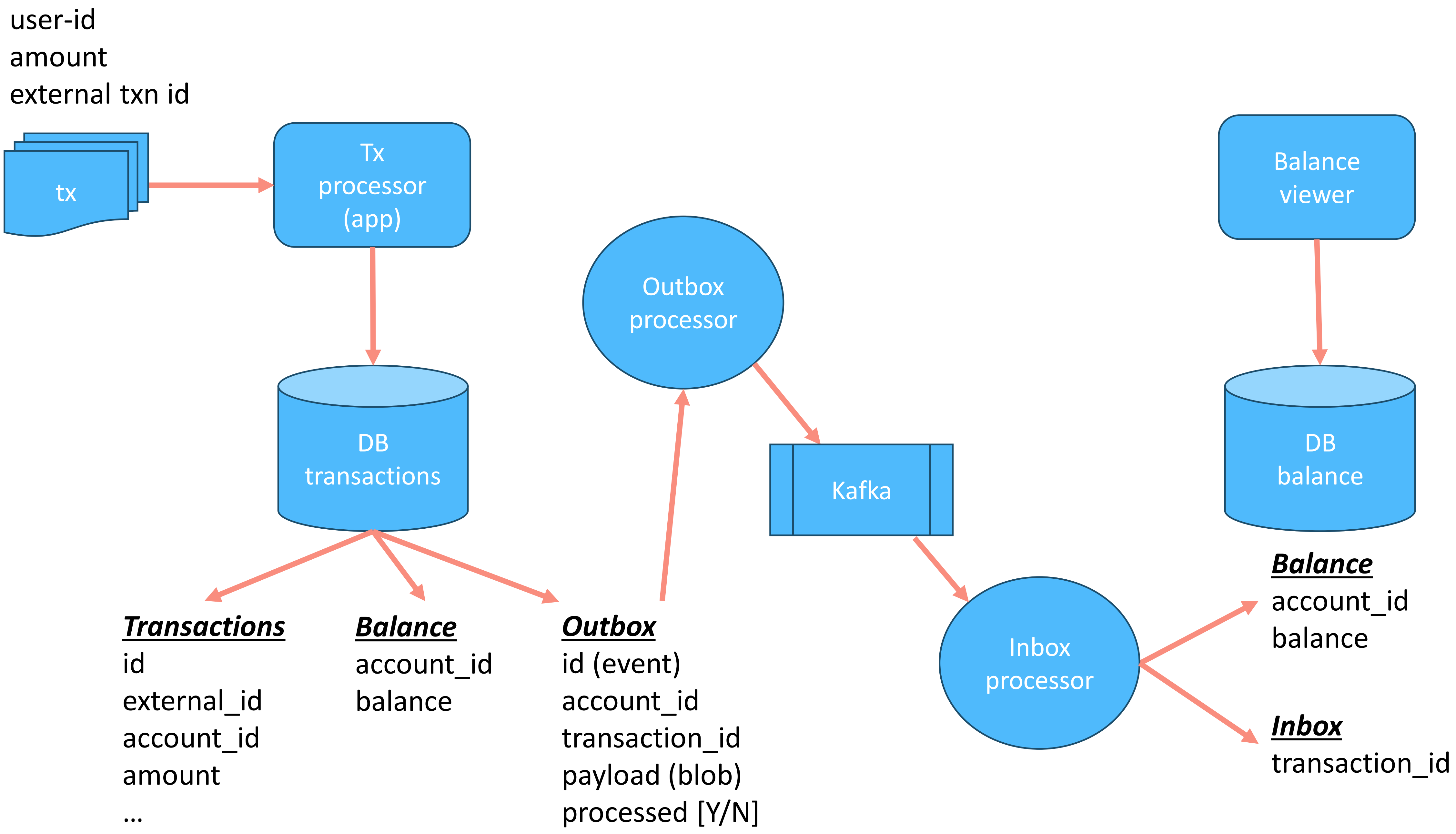
Homework

Implement consistent transaction replication flow

- Deliver transactions from one database to another
- Maintain consistency in the face of every component outage
- Use queue or implement direct outbox-to-inbox

Demo


user-id
amount
external txn id
tx
Tx processor (app)
DB transactions
Transactions
id
external_id
account_id
amount
…
Balance
account_id
balance
Balance viewer
DB balance
Balance
account_id
balance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balance
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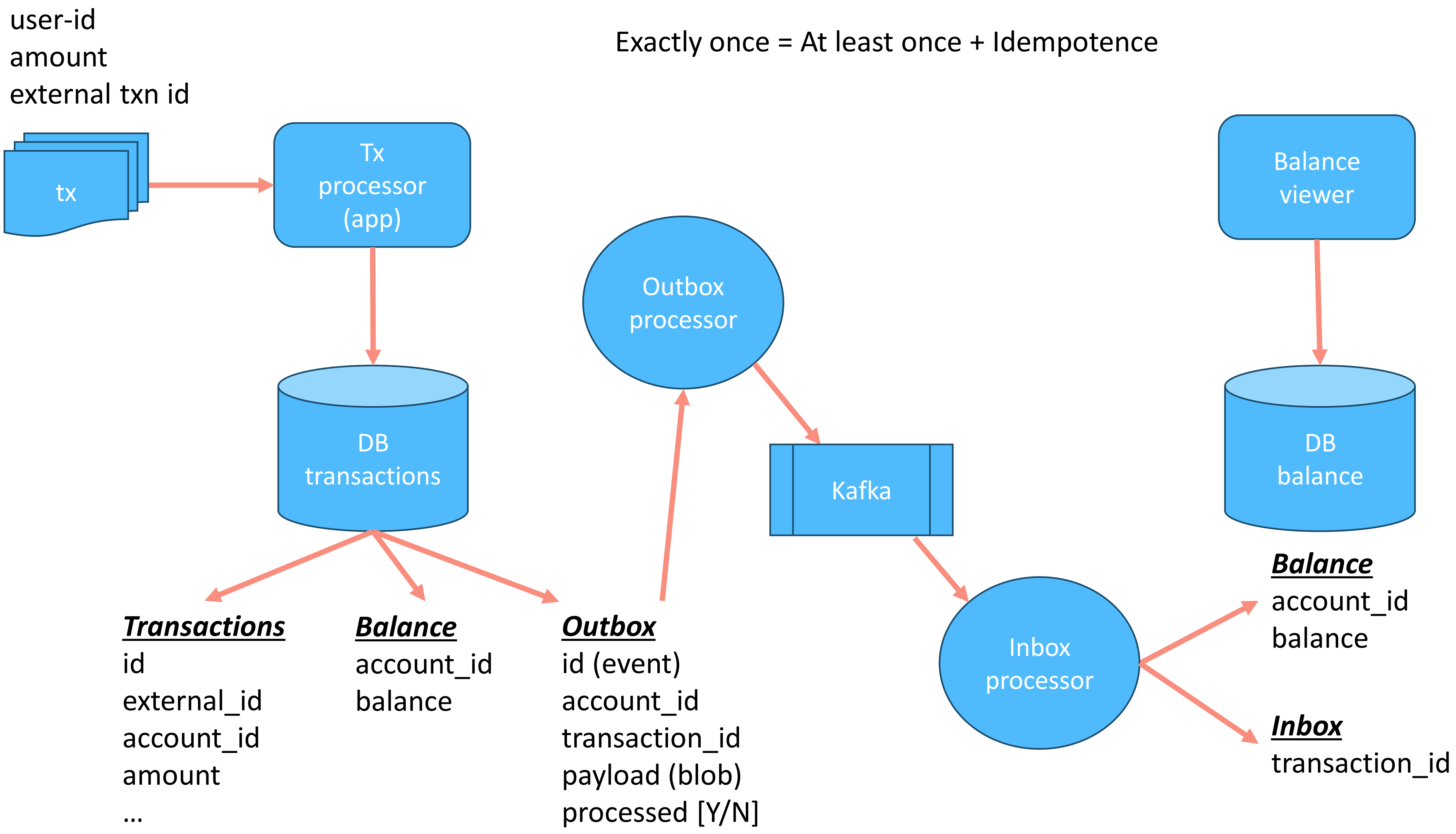
external_id
account_id
amount
...
Balance
account_id
balance
Outbox
id (event)
account_id
transaction_id
payload (blob)
processed [Y/N]
Inbox
processor
Balance
account_id
balance
Inbox
transaction_id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Exactly once = At least once + Idempotence
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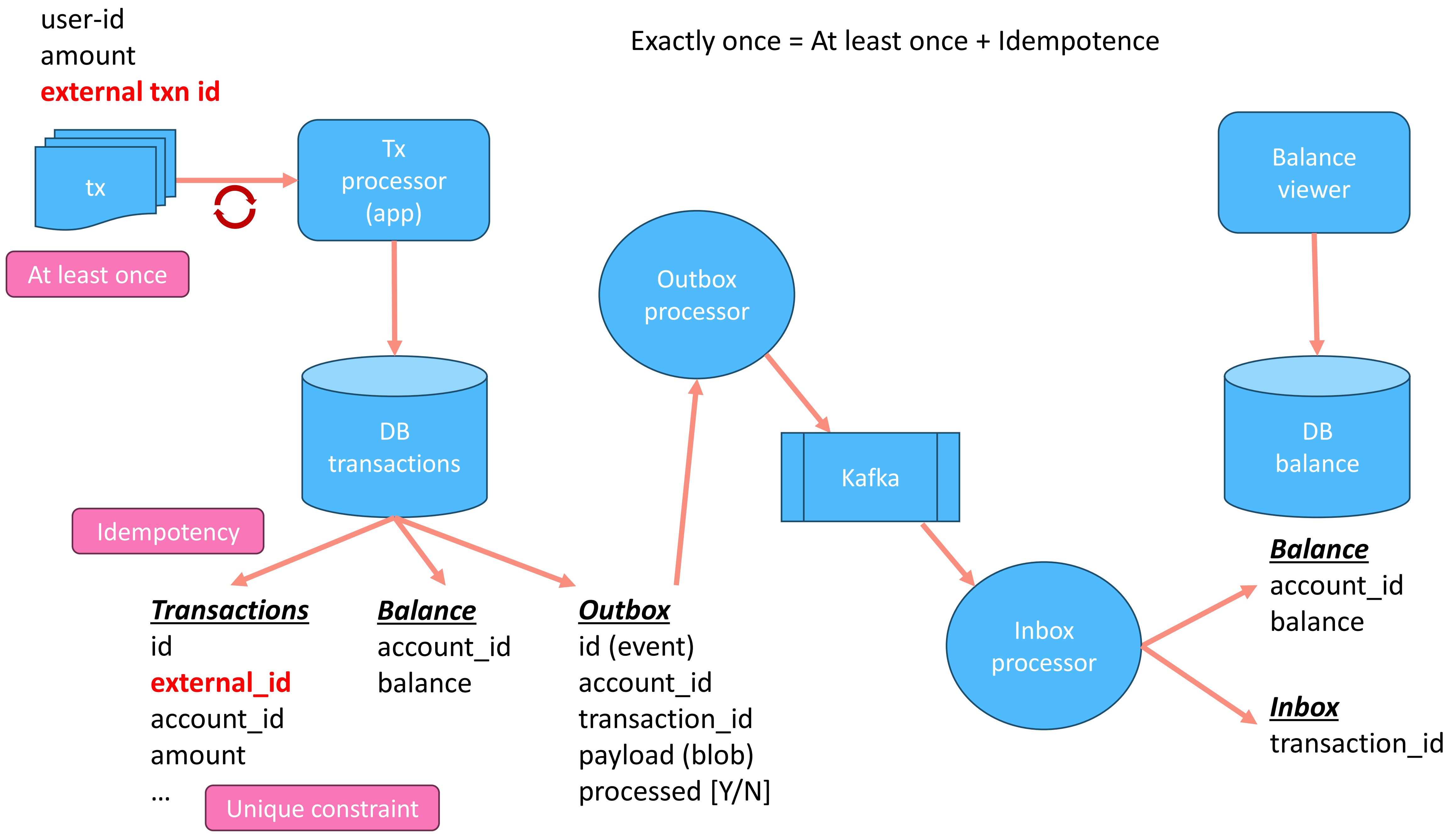
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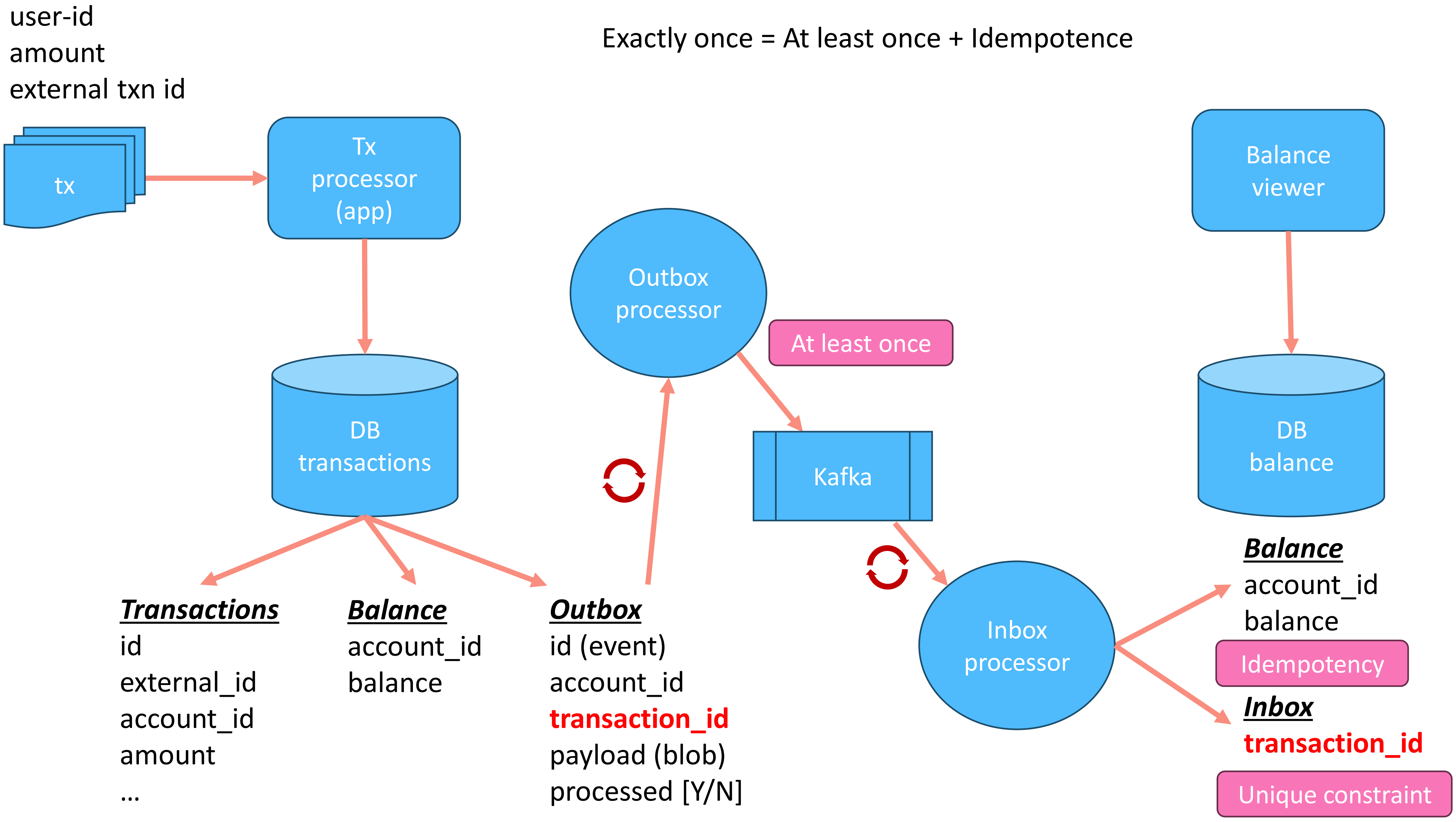
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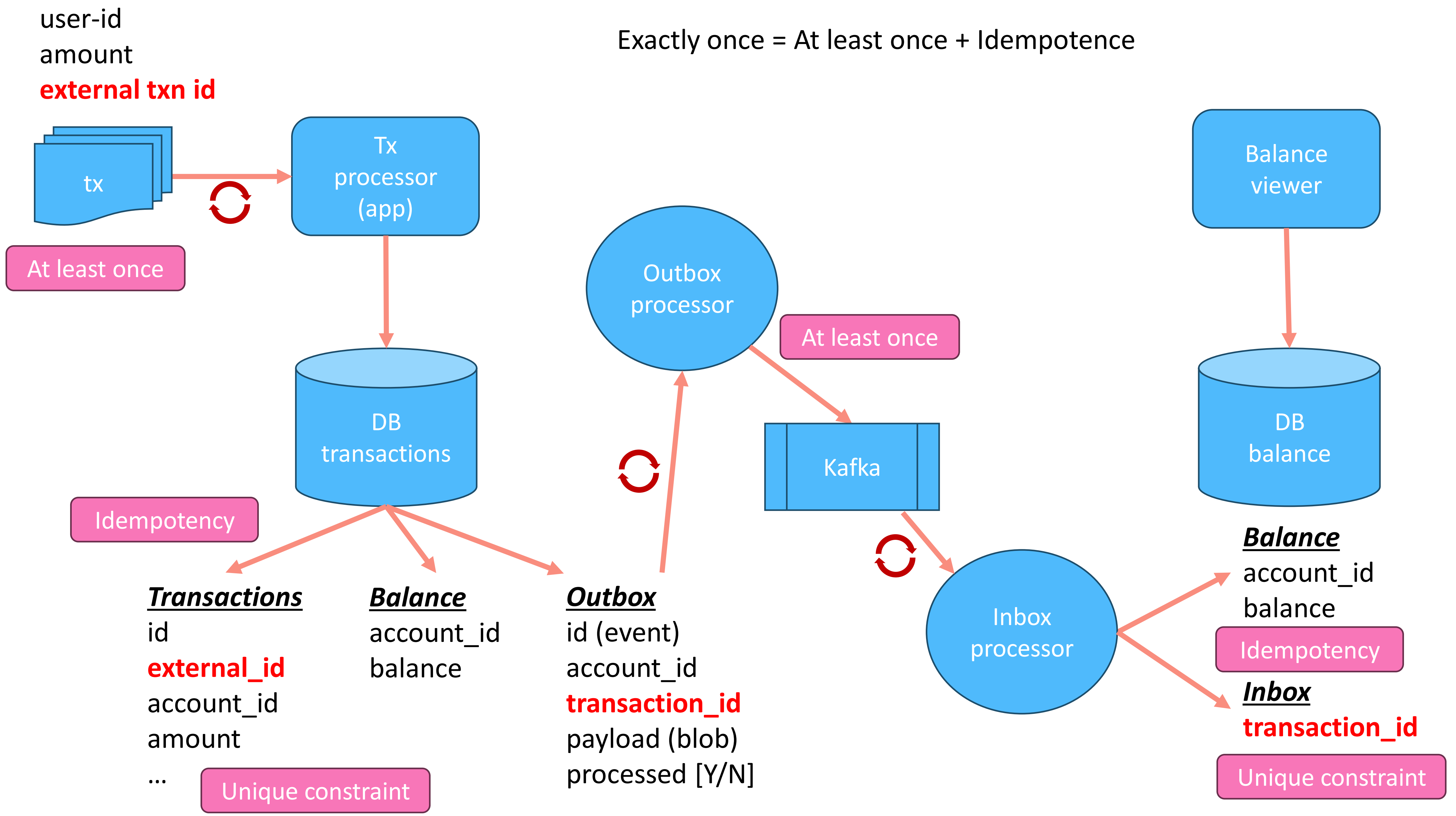
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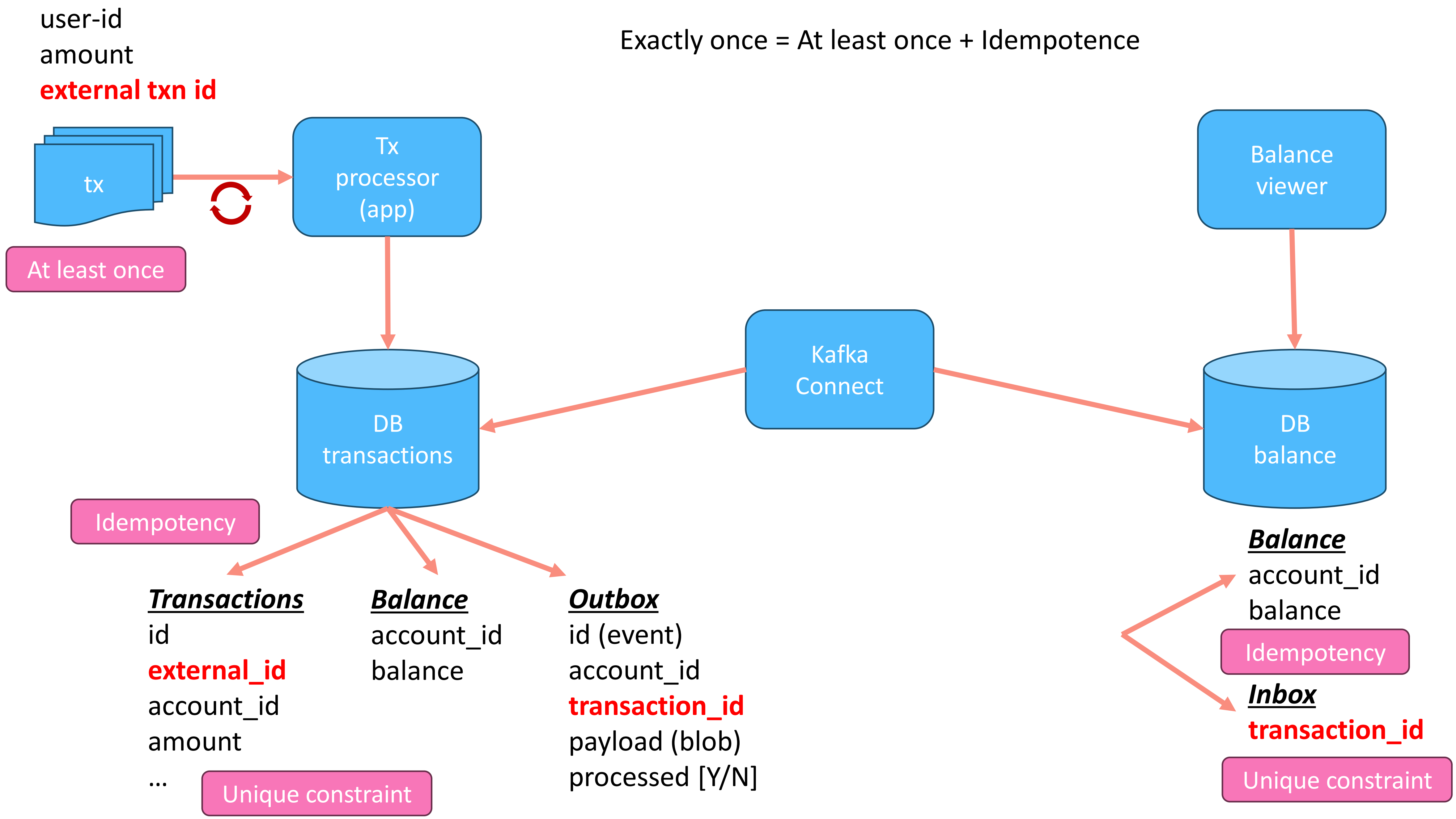

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Kafka Connect
DB transactions
DB balance
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